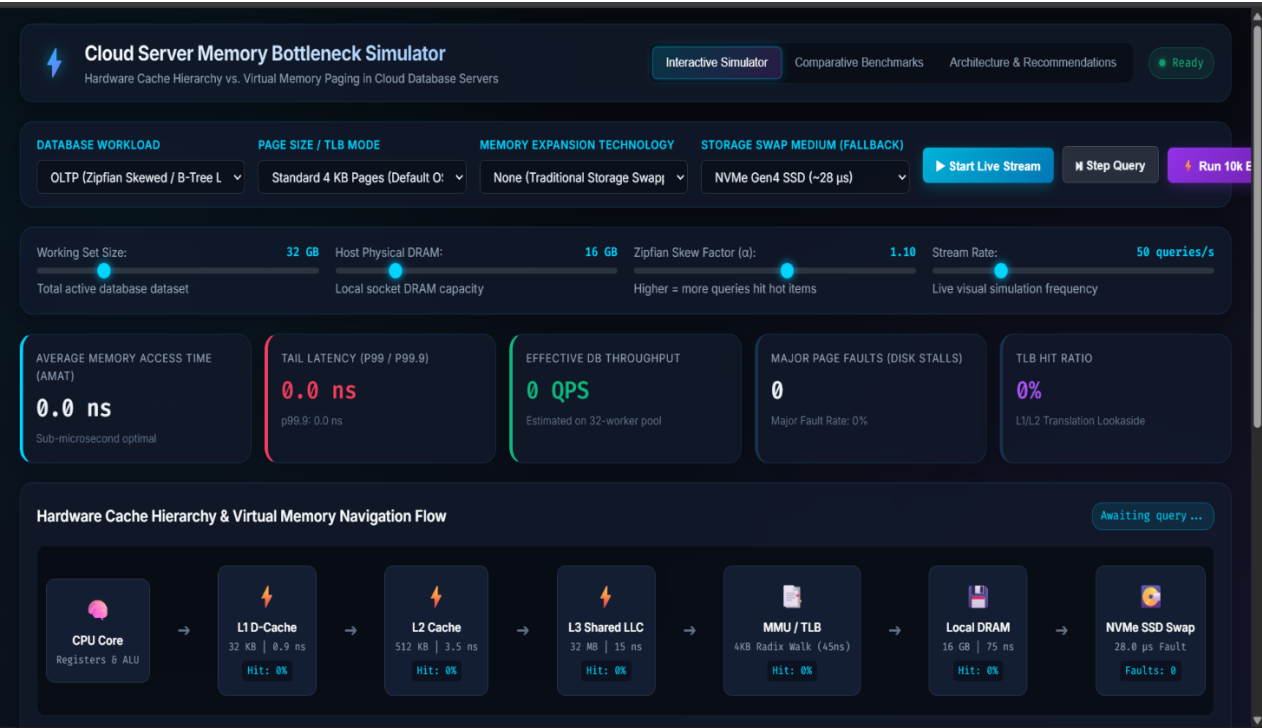


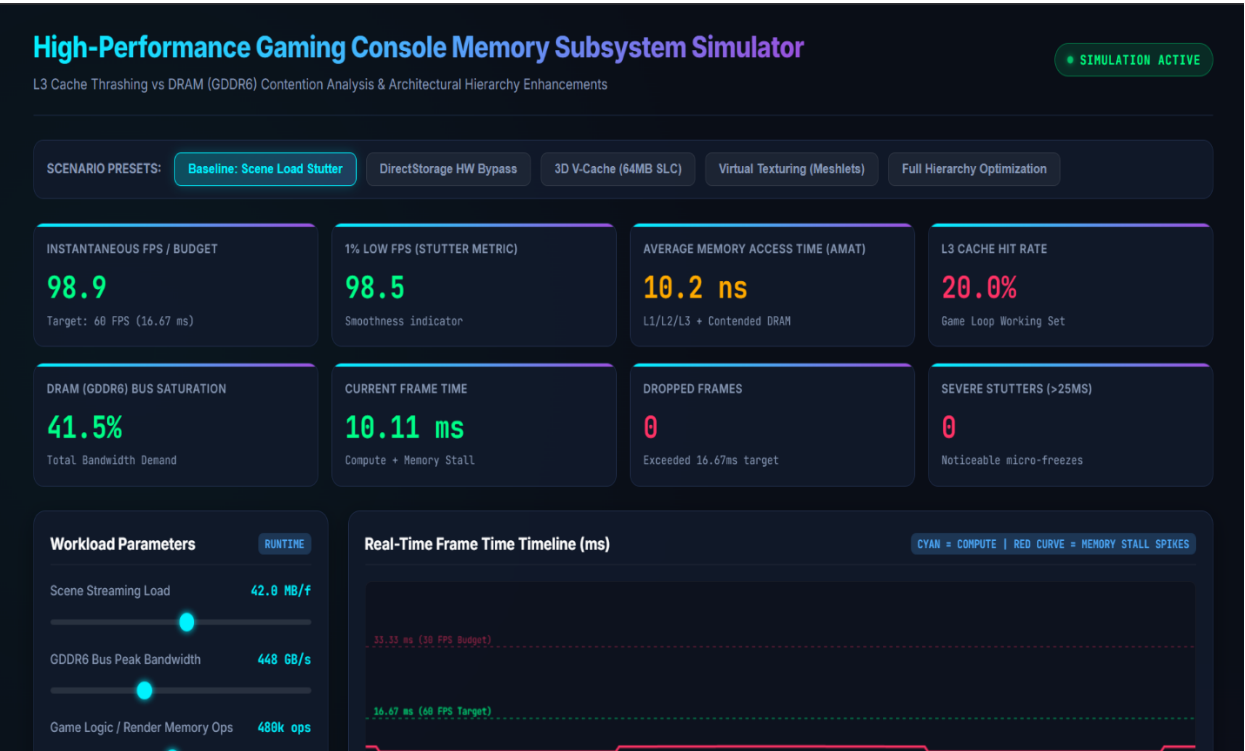
1.CLOUD SERVER BOTTLENECK



2.EDGE AI MEMORY HIERARCHY



3.GAMING CONSOLE



4.SMART MOBILE HIERARCHY

SMARTPHONE MEMORY HIERARCHY APP-SWITCHING SIMULATION RESULTS			
Performance / Subsystem Metric	Baseline SoC	Proposed Redesign	Improvement
Cold App Switch Latency (Cycles)	3,512,341	3,045,598	-13.3% Latency
Warm App Resume Latency (Cycles)	538,338	490,358	-8.9% Latency
Total Benchmark Latency (Cycles)	9,243,419	8,080,718	-12.6% Cycles
Average Memory Access Time (AMAT)	117.01 cycles	102.29 cycles	-12.6% Faster
L1 Data Cache Hit Rate	4.70%	8.90%	+4.2% pts
L1 Instruction Cache Hit Rate	4.82%	4.96%	+0.1% pts
L2 Private Cache Hit Rate	49.48%	54.52%	+5.0% pts
L3 Shared Cluster Cache Hit Rate	22.58%	13.04%	Filtered by L2
System-Level Cache (SLC) Hit Rate	0.01%	0.00%	Filtered by L3
Total DRAM Accesses (Demand + WB)	29,424	29,422	Stable Count
DRAM Row Buffer Hit Rate	3.29%	3.46%	+0.2% pts
DRAM Row Conflicts	28,441	28,388	-0.2%
Total Memory Subsystem Energy	145.19 uJ	144.23 uJ	-0.7% Energy

5.VEHICLE MEMORY HIERARCHY

